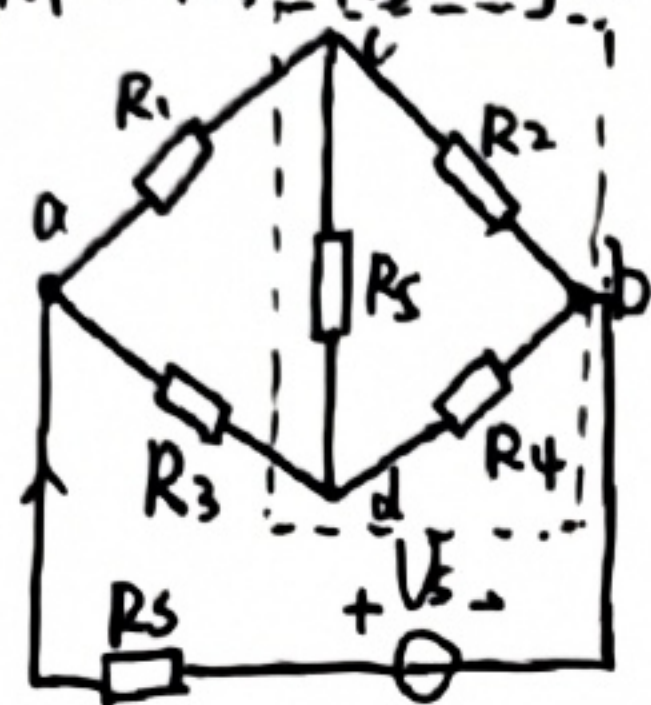
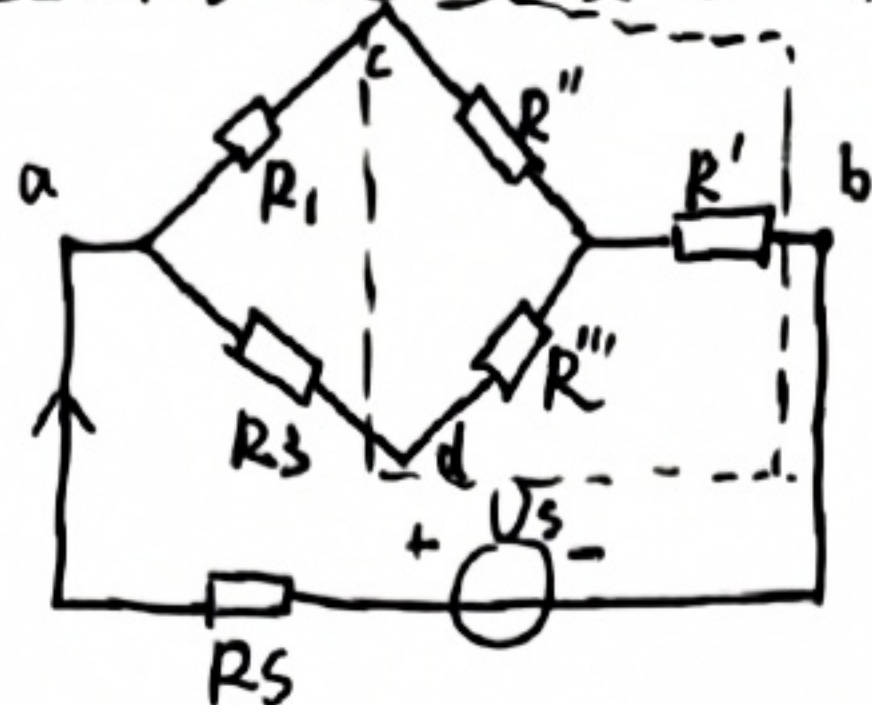


2. $R_1 = R_4 = R_5 = 4\Omega$, $R_2 = 8\Omega$, $R_3 = 5\Omega$, $R_6 = 5\Omega$, $U_S = 12V$. 求 I 及流过 R_1 的电流 I_1 .



$\Rightarrow$



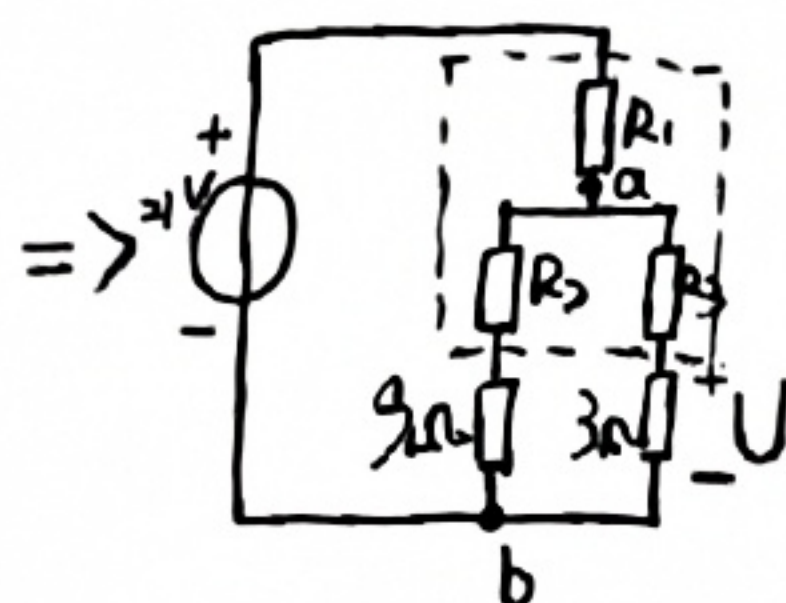
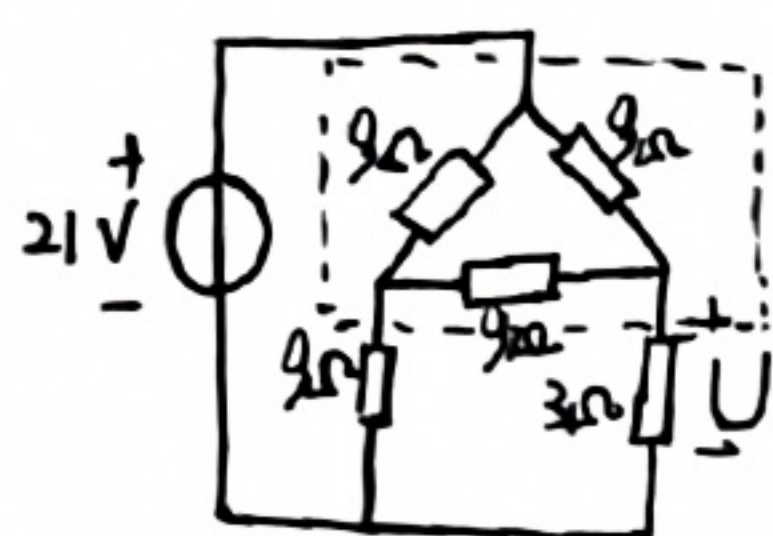
$$R' = \frac{R_2 \cdot R_4}{R_2 + R_4 + R_5} = 2\Omega$$

$$R'' = \frac{R_2 \cdot R_5}{R_2 + R_4 + R_5} = 2\Omega$$

$$R''' = \frac{R_4 \cdot R_5}{R_2 + R_4 + R_5} = 1\Omega$$

$$R_{eq} = R_5 + [(R_1 + R'') // (R_3 + R''') + R'] = 10\Omega \quad I = \frac{U_S}{R_{eq}} = 1.2A \quad I_1 = I \cdot (R_{eq} - R_5 - R') / (R_1 + R'') = 0.6A$$

3. 电阻的Y-Δ等效变换求 U .



$$R_1 = \frac{9 \times 9}{9+9+9} \Omega = 3\Omega$$

$$R_2 = \frac{9 \times 9}{9+9+9} \Omega = 3\Omega$$

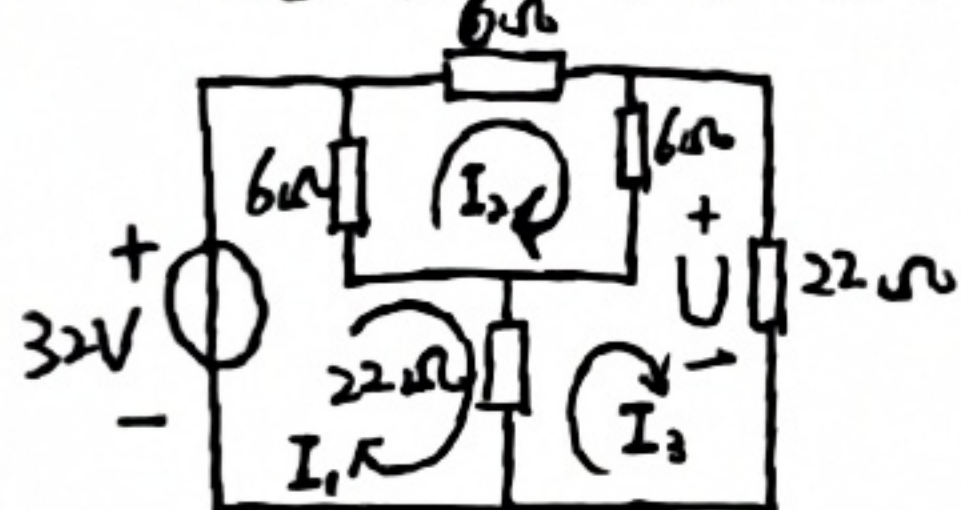
$$R_3 = \frac{9 \times 9}{9+9+9} \Omega = 3\Omega$$

$$R_{eq} = R_1 + [(R_2 + R_3) // (R_4 + R_5)] = 7\Omega$$

$$U_{ab} = 24 - 24 \times \frac{R_1}{R_{eq}} = 12V$$

$$U = U_{ab} \times \frac{3}{3+R_3} = 6V \text{ --- 分压原理}$$

4. 网孔电流法求 U .



如图, 确定网孔电流方向.

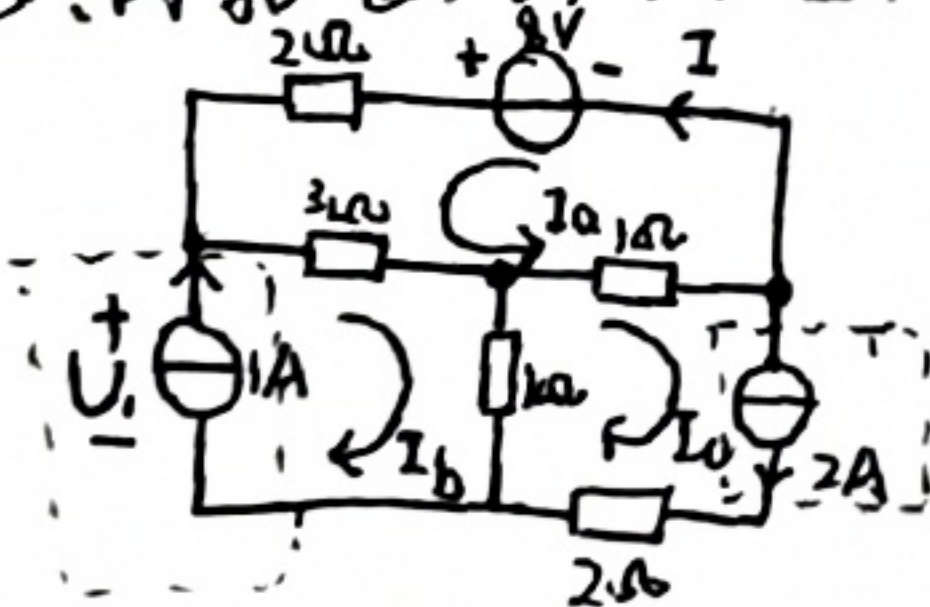
$$\textcircled{1} \text{ 第1个网孔: } (6+22) \times I_1 - 6I_2 - 22I_3 = 32$$

$$\textcircled{2} \text{ 第2个网孔: } (6+6+6) \times I_2 - 6I_1 - 6I_3 = 0$$

$$\textcircled{3} \text{ 第3个网孔: } (6+22+22) \times I_3 - 22I_1 - 6I_2 = 0$$

$$\Rightarrow \begin{cases} I_1 = \frac{16}{7}A \\ I_2 = \frac{2}{7}A \\ I_3 = \frac{8}{7}A \end{cases} \quad U = I_3 \cdot 22V = 25.14V$$

5. 网孔电流法求 I .



由图可知, 一般电流方向定为独立电源正方向, 方便计算.

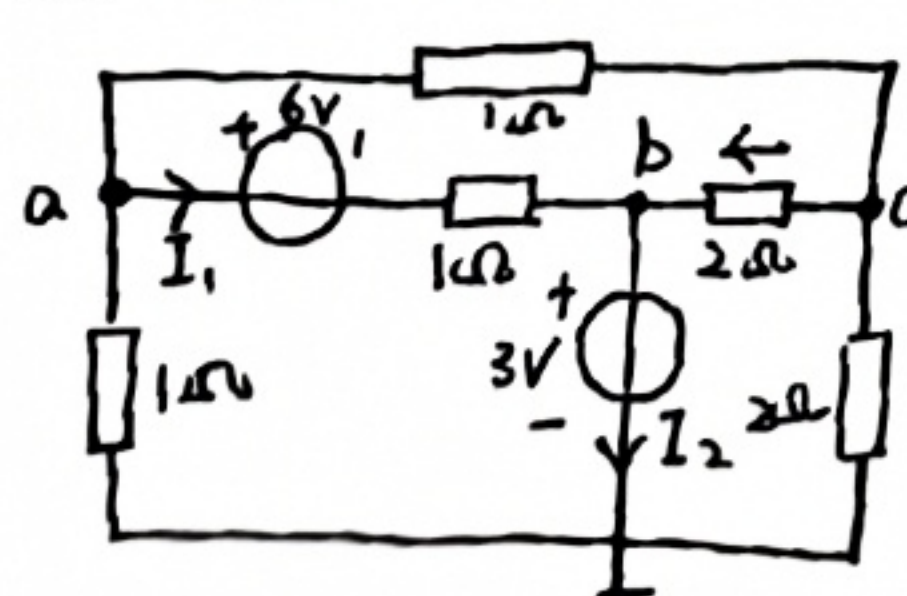
$$I_b = 1A \text{ (独立支路有唯一电流源)}$$

$$I_c = 2A \text{ (独立支路有唯一电流源)}$$

$$\text{可得 } I = I_a = 0.5A$$

$$\text{a 环路: } (2+3+1) \cdot I_a + 3I_b + 1 \times I_c = 8$$

6. 结点电压法求各结点电压及电流 I_1 和 I_2 .



如图, a, b, c 为电路图三个结点.

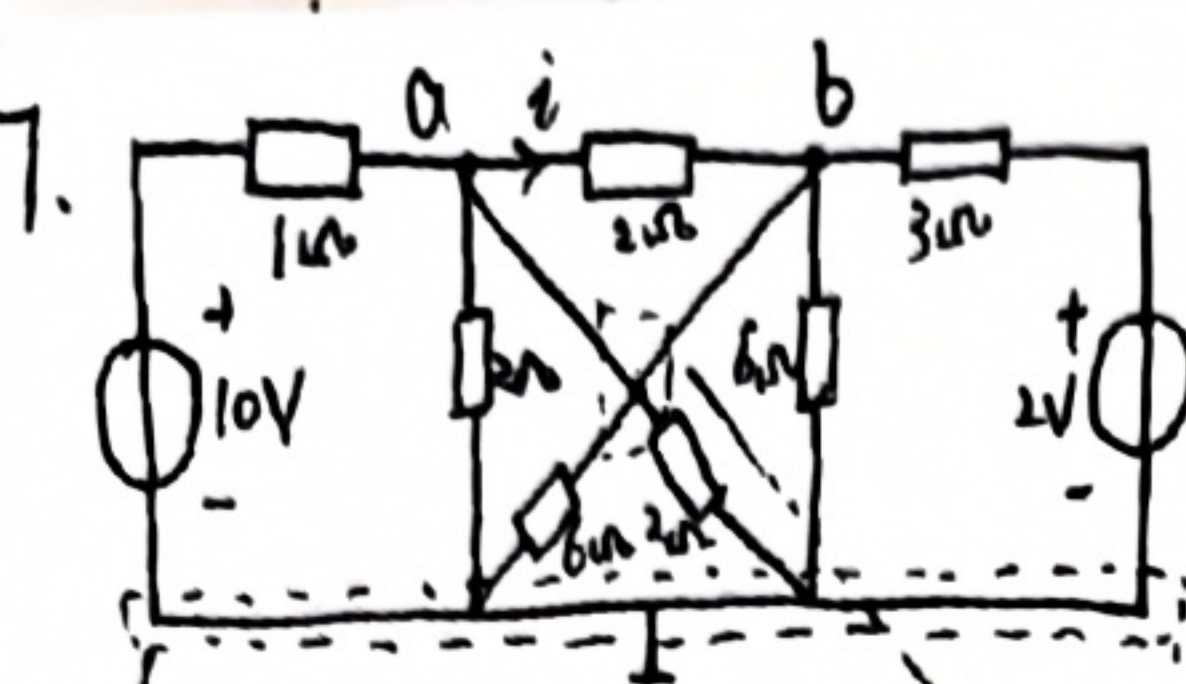
$$a: (\frac{1}{1} + \frac{1}{1} + \frac{1}{1}) U_a - \frac{1}{1} \times U_b - \frac{1}{1} \times U_c = \frac{6}{1}$$

$$b: \text{结点 b 与零电位点相连只有唯一电压源} \Rightarrow \begin{cases} U_a = 3.9V \\ U_b = 3V \\ U_c = 2.7V \end{cases}$$

$$c: (\frac{1}{1} + \frac{1}{2} + \frac{1}{2}) U_c - \frac{1}{2} \times U_b - \frac{1}{1} \times U_a = 0$$

$$U_{ab} = U_a - U_b = 0.9V \quad I_1 = (U_{ab} - 6) / 1 = -5.1A$$

$$\text{定义电流方向 } c \rightarrow b: U_{cb} = U_c - U_b = -0.3V \quad I_2 = U_{cb} / 2 + I_a = -5.25A$$



结点电压求*i*.

$$a: (\frac{1}{1} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}) U_a - \frac{1}{2} U_b = \frac{10}{1}$$

$$b: (\frac{1}{3} + \frac{1}{6} + \frac{1}{6} + \frac{1}{2}) \cdot U_b - \frac{1}{2} U_a = \frac{2}{3}$$

符号: 流入结点为正, 流出为负.

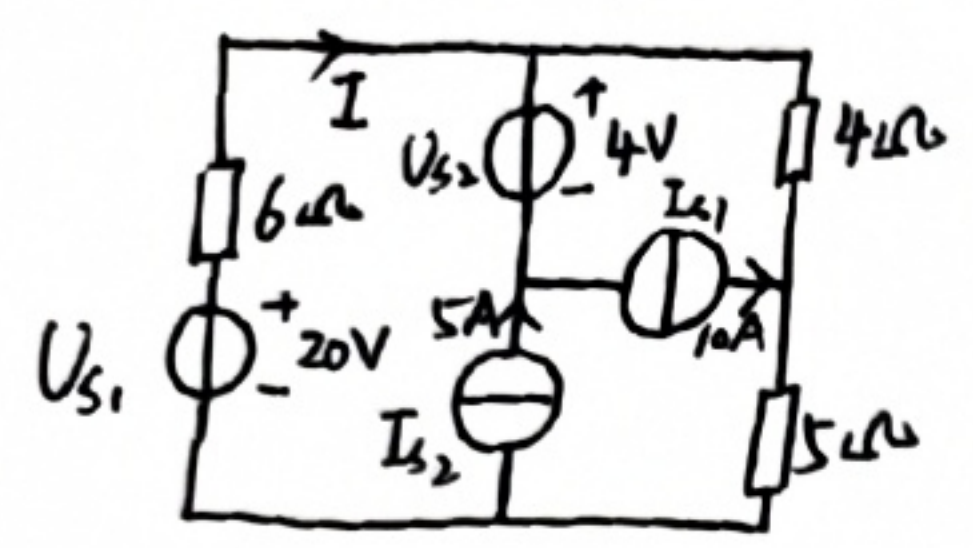
$$\Rightarrow \begin{cases} U_a = 4.5V \\ U_b = 2.5V \end{cases} \quad \begin{aligned} a \rightarrow b \\ i = \frac{U_a - U_b}{2} A \\ = 1A. \end{aligned}$$

为同一个电位点

① 取参考电位点 → 此点不相交, 只为导线互搭, 相交的点会加粗.

8. 叠加定理, 求*I*.

每个理想电源元件单独作用后叠加不作用的电源置0. 即电流源开路, 电压源短路.



①. U_{S1} : $I_1 = \frac{20}{6+4+5} A = \frac{4}{3} A$

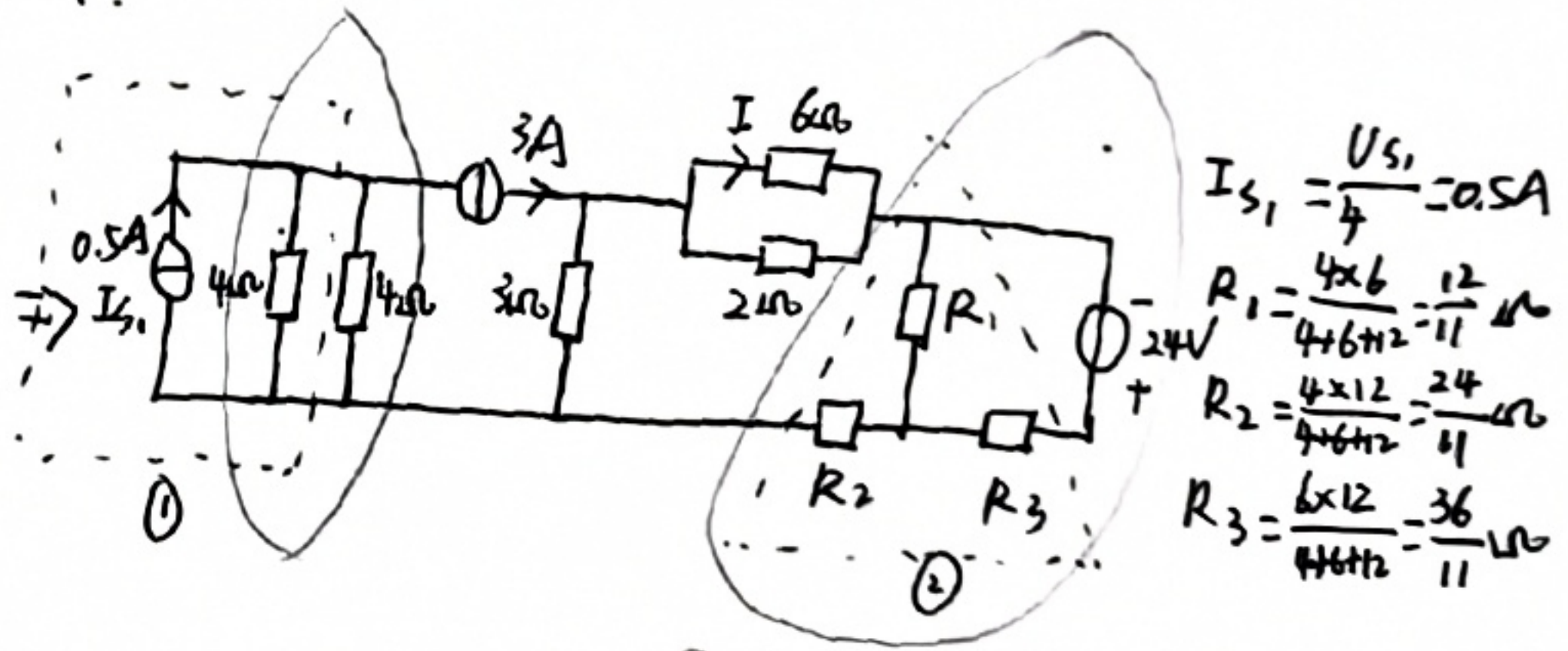
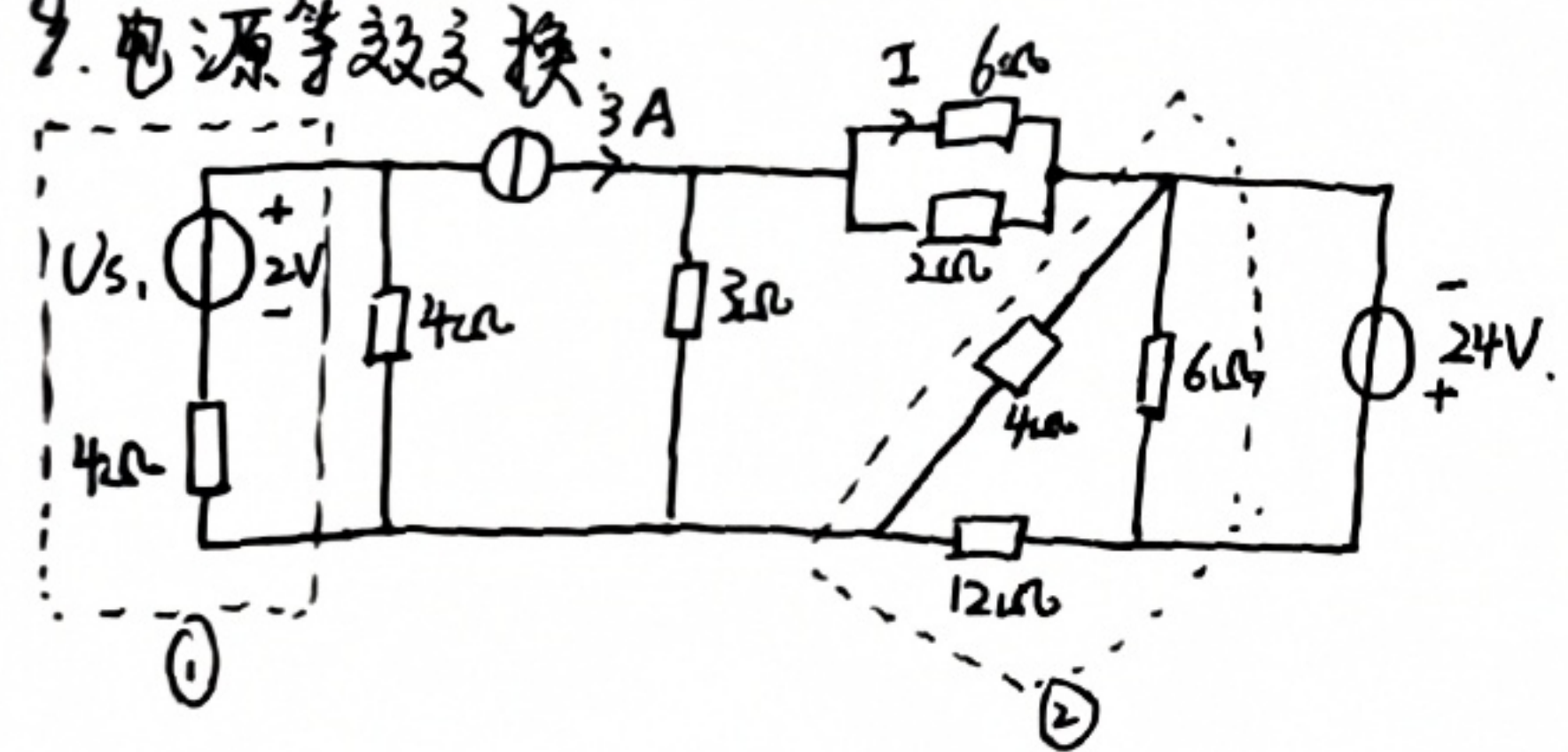
②. U_{S2} : (X) 不在回路中作用.

③. I_{S1} : 分流公式: $I_3 = \frac{4}{4+5+6} \times 10 A = \frac{2}{3} A$

④. I_{S2} : 分流公式: $I_4 = -\frac{4+5}{(4+5)+6} \times 5 = -3A$

$I = I_1 + I_3 + I_4 = 1A$

9. 电源等效变换:

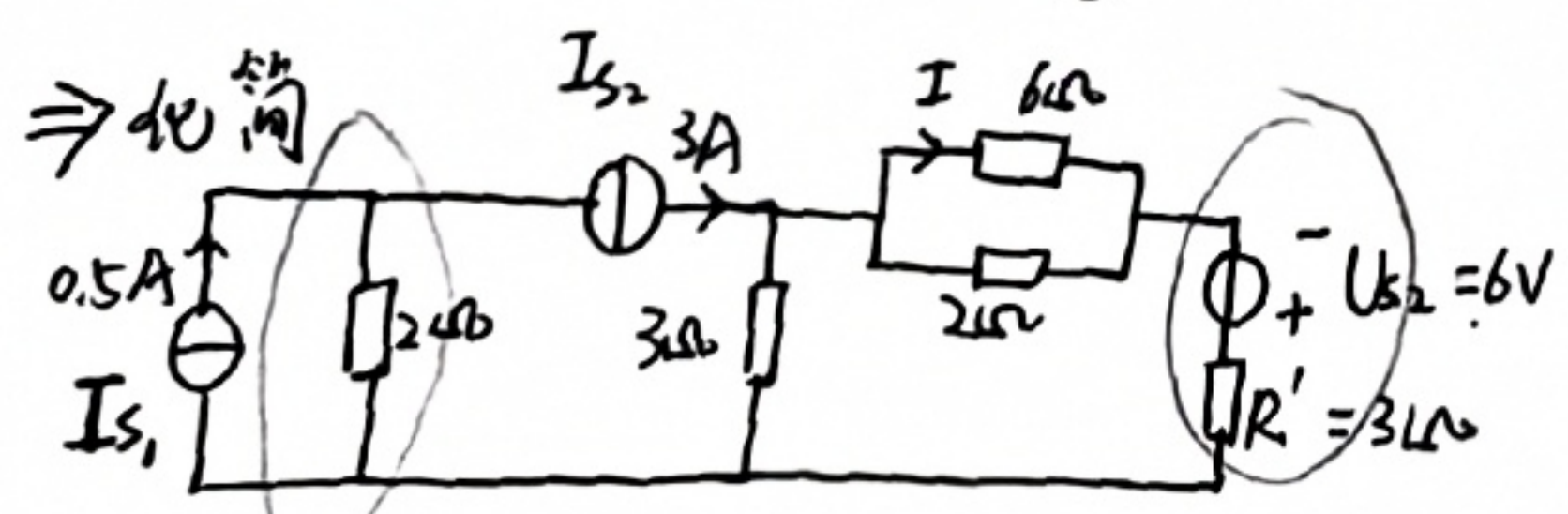


$$I_{S1} = \frac{U_{S1}}{4} = 0.5A$$

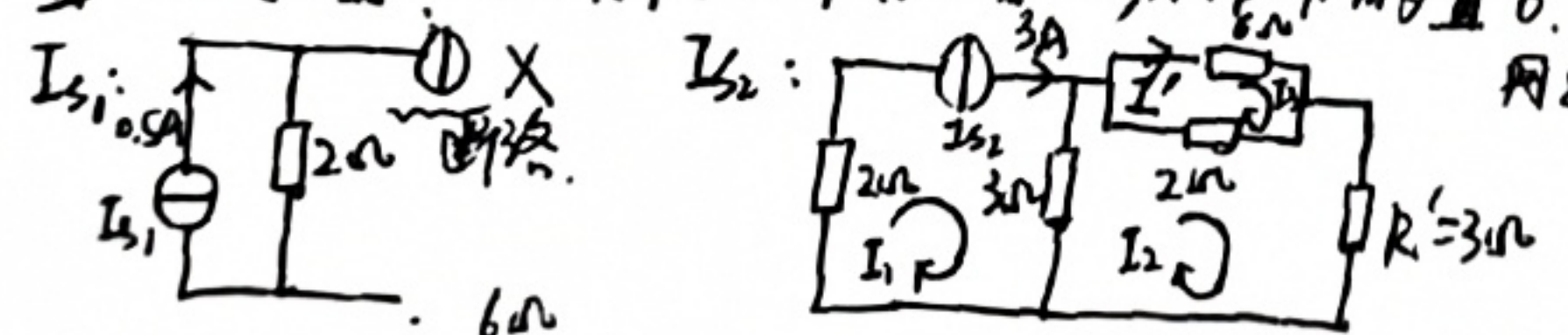
$$R_1 = \frac{4 \times 6}{4+6} = \frac{12}{5} \Omega$$

$$R_2 = \frac{4 \times 12}{4+12} = \frac{24}{4} \Omega$$

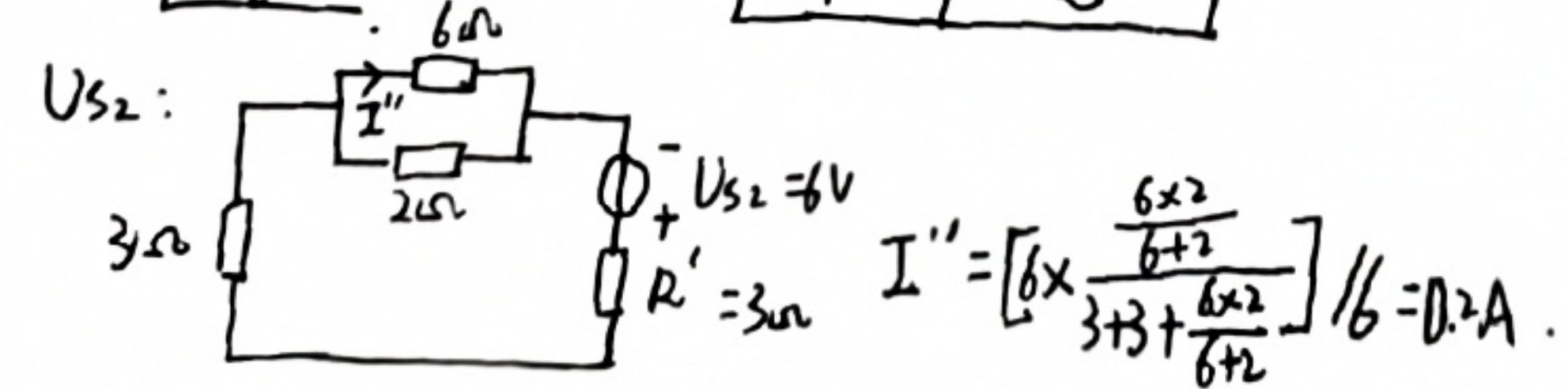
$$R_3 = \frac{6 \times 12}{6+12} = \frac{36}{11} \Omega$$



叠加定理 电源单独作用后叠加, 不作用的置0.



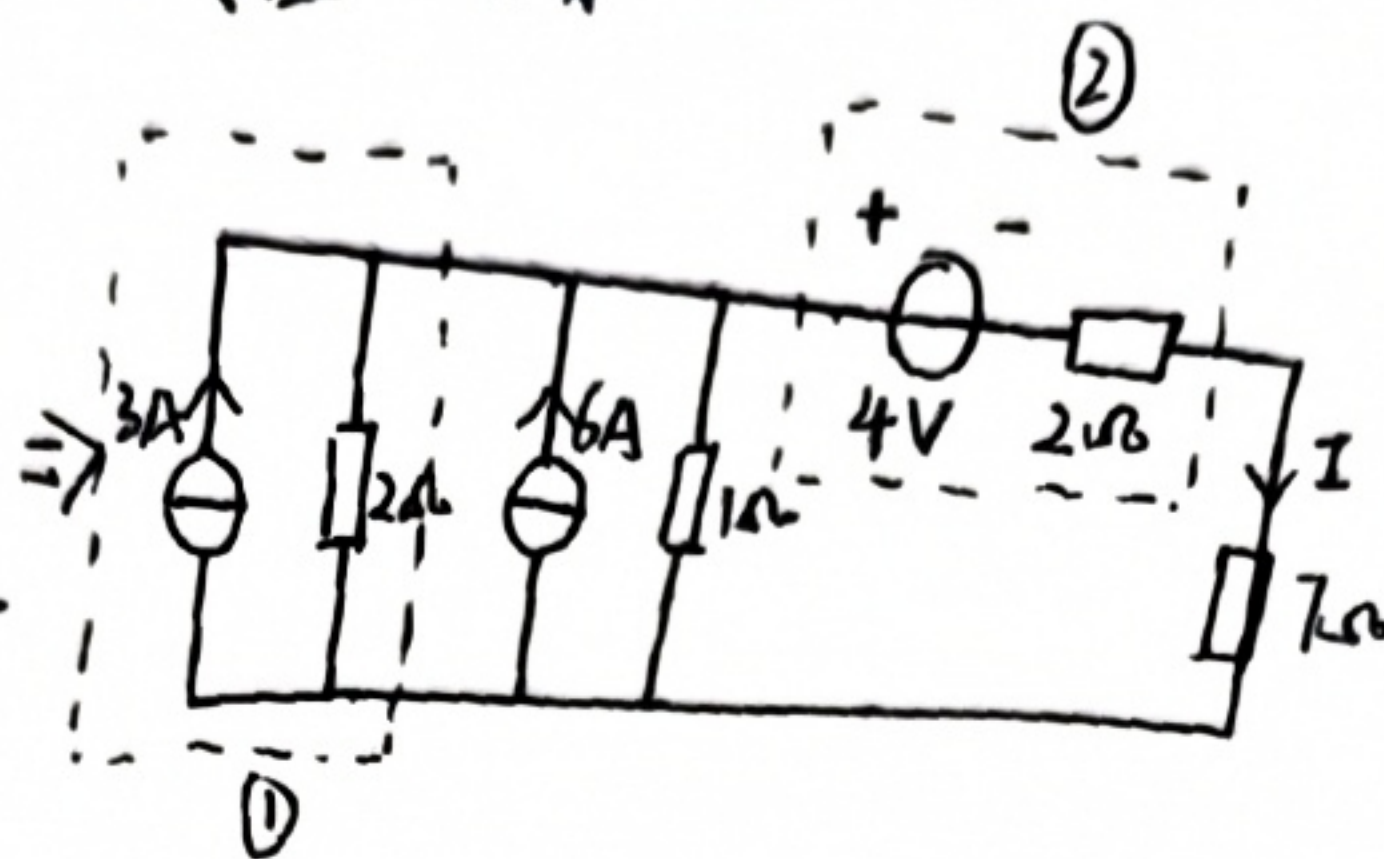
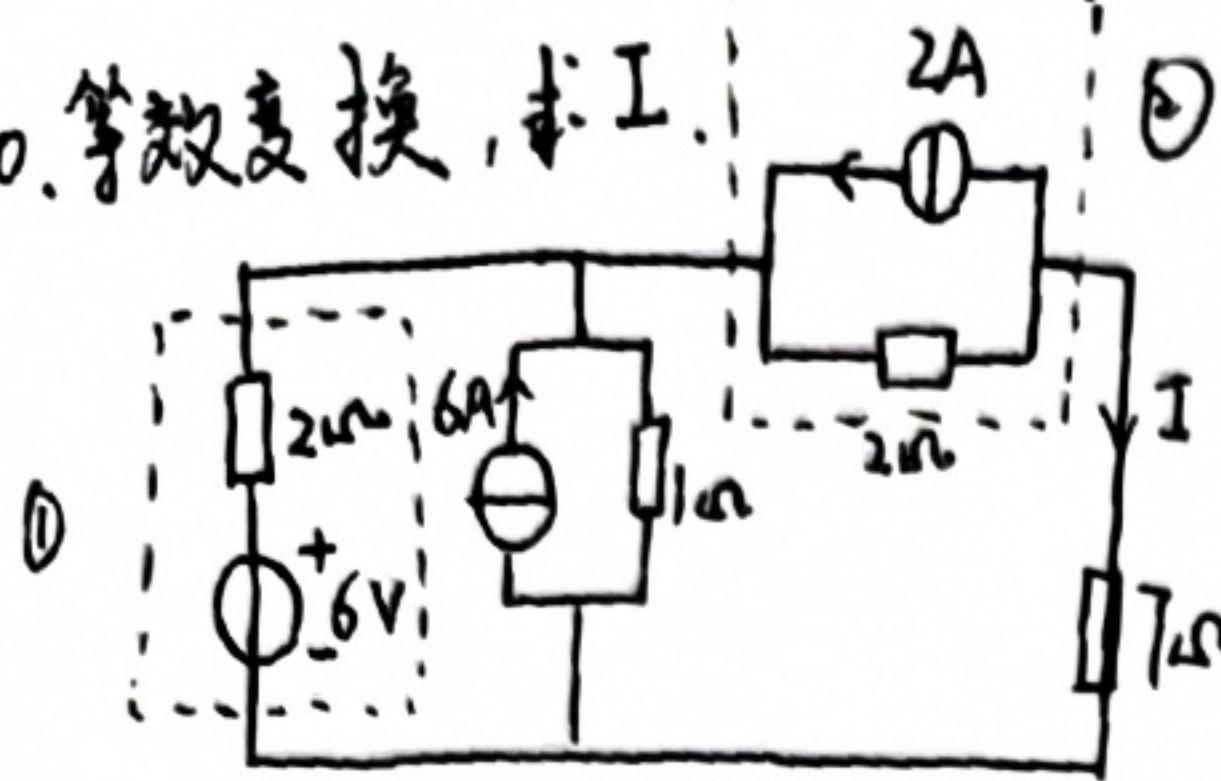
网孔电流法: 1: $I_1 = 3A$
 2: $(3+2+3) \cdot I_2 - 3 \cdot I_1 - 2 \cdot I_3 = 0 \Rightarrow \begin{cases} I_1 = 3A \\ I_2 = 1.2A \\ I_3 = 0.3A \end{cases}$
 3: $(6+2) \cdot I_3 - 2 \cdot I_2 = 0$
 3P $I' = I_3$



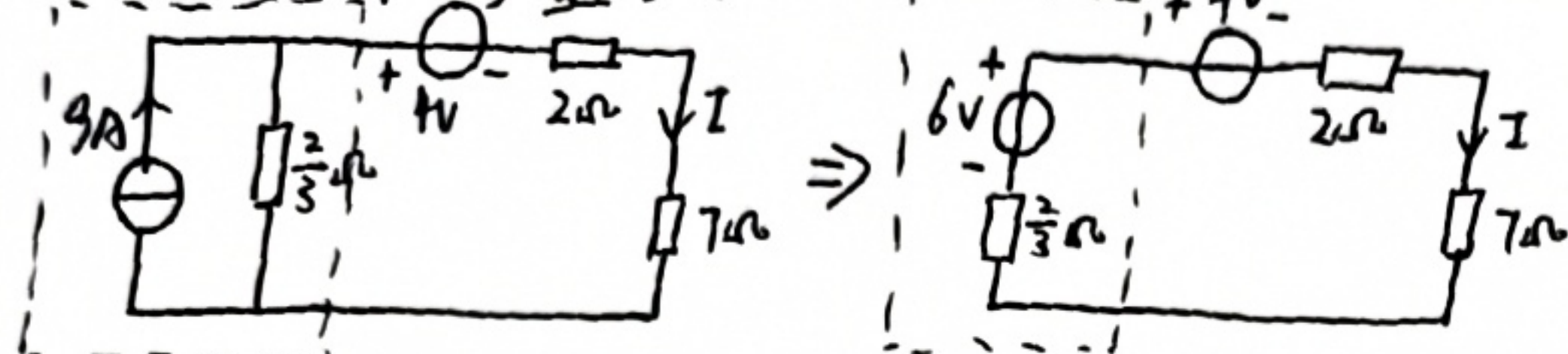
$I = I' + I'' = 0.5A$

(注: 将最左为 8A 的电流源删除, 原答案在附录页)

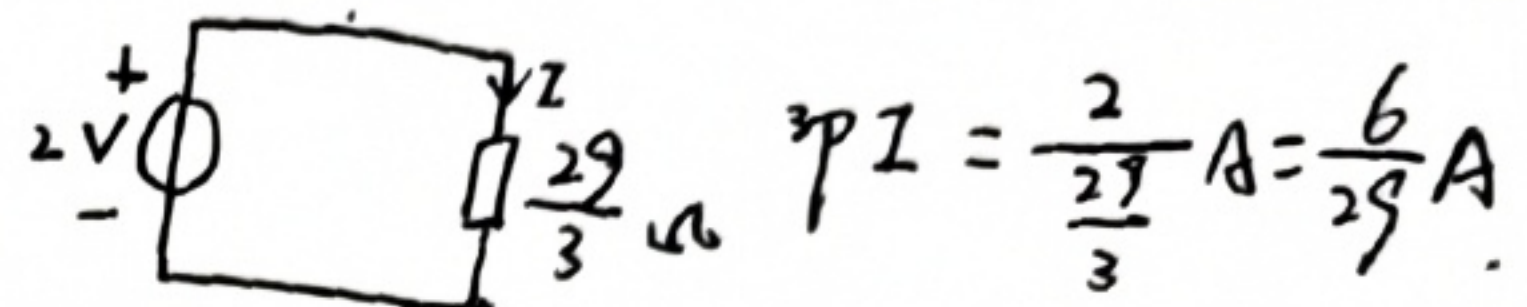
10. 等效变换, 求 I .



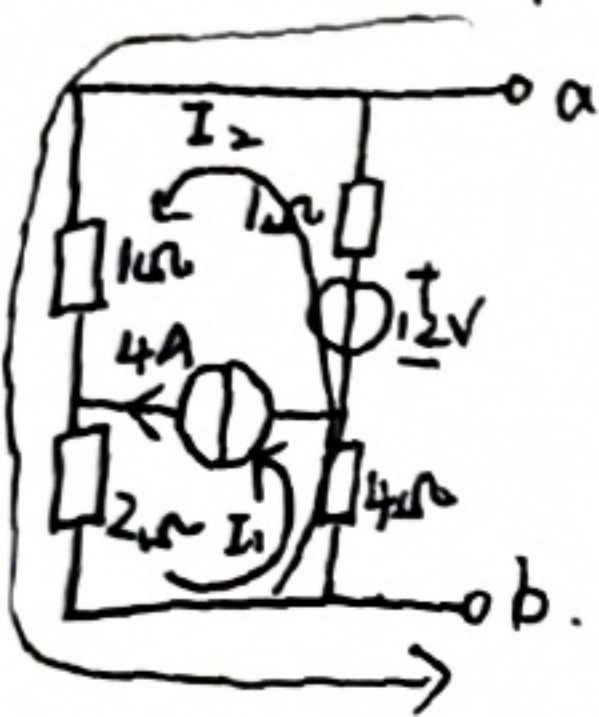
电流源并联可叠加:



电压源串联可叠加:



11. 求戴维南等效电路:



开路电压 U_{oc}
用网孔电流法求解.

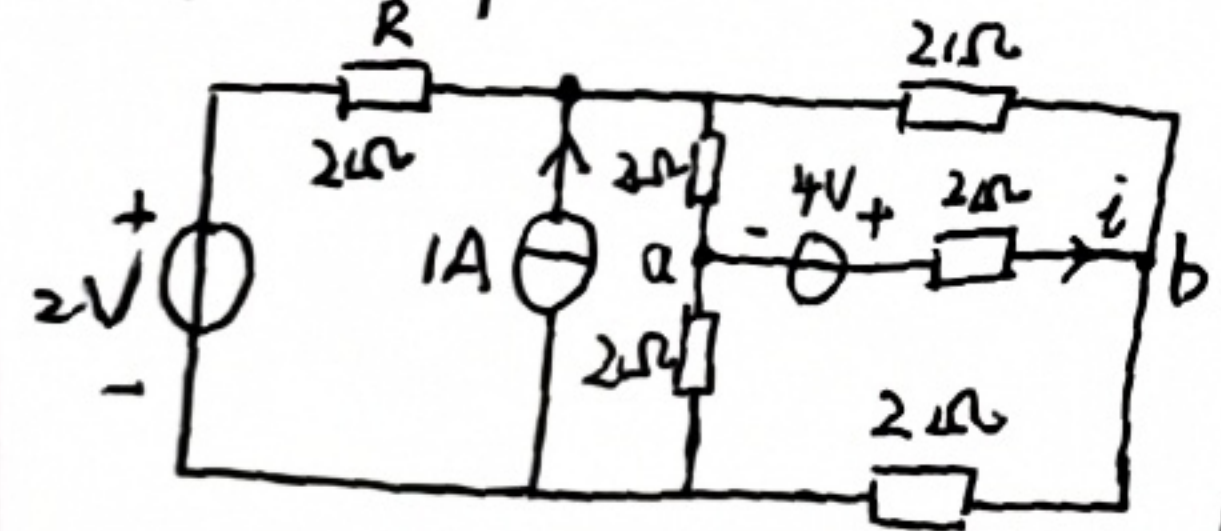
1. $I_1 = 4A$.

2. $(1+1+2+4)I_2 + (2+4)I_1 = 12 \Rightarrow \begin{cases} I_1 = 4A \\ I_2 = -1.5A \end{cases}$

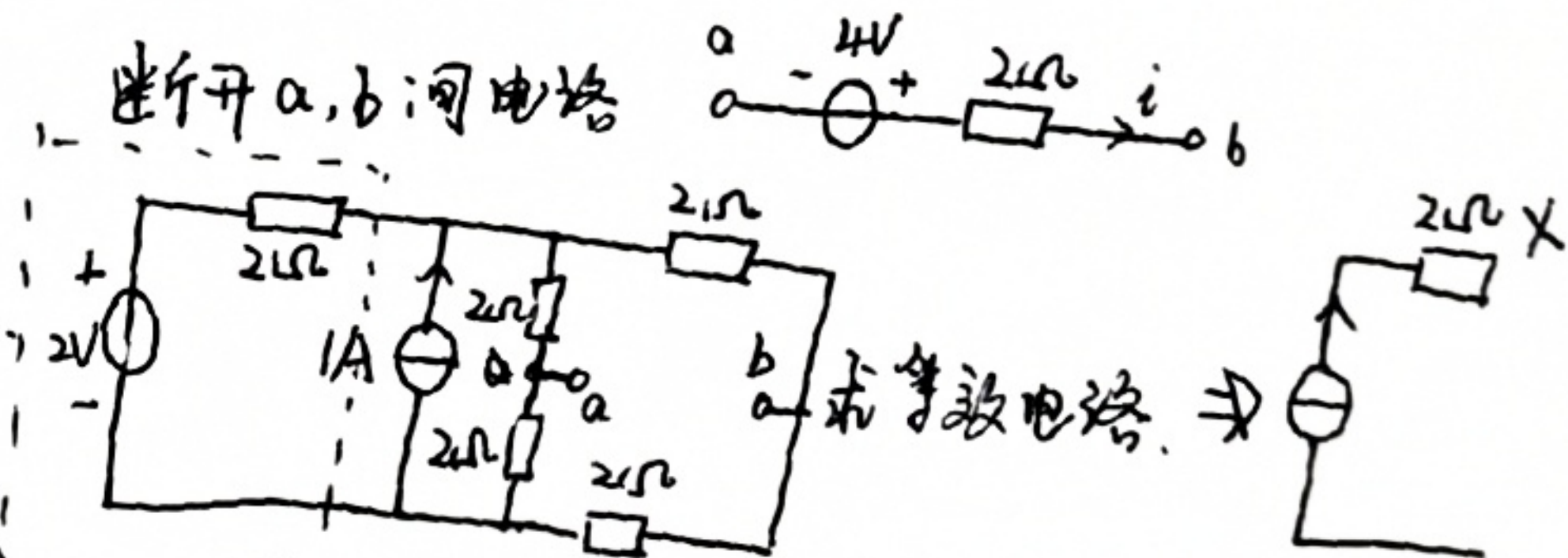
外围点格: KVL 求解.

$U_{oc} = (1+2) \times I_3 + 2 \times I_1 = 3.5V$.

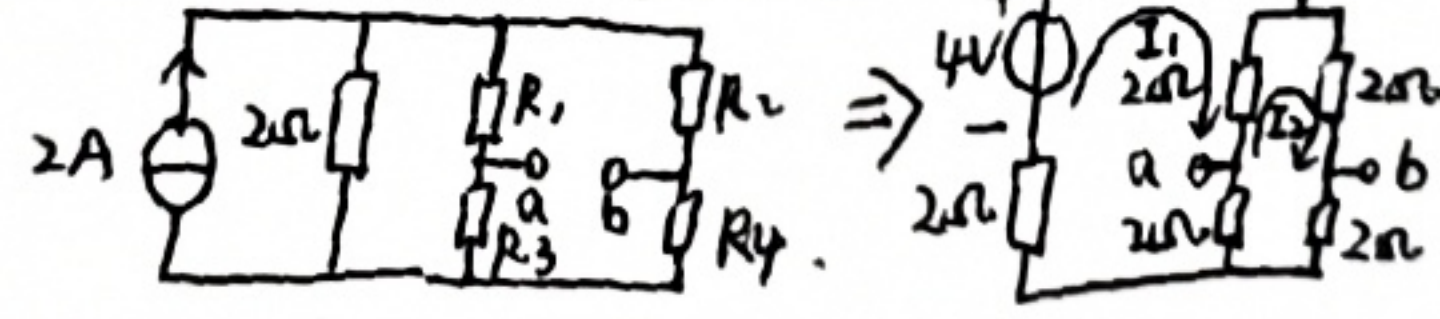
12. 戴维南定理 $\rightarrow i$.



断开 a, b 间电路



令电流源叠加 (并联)



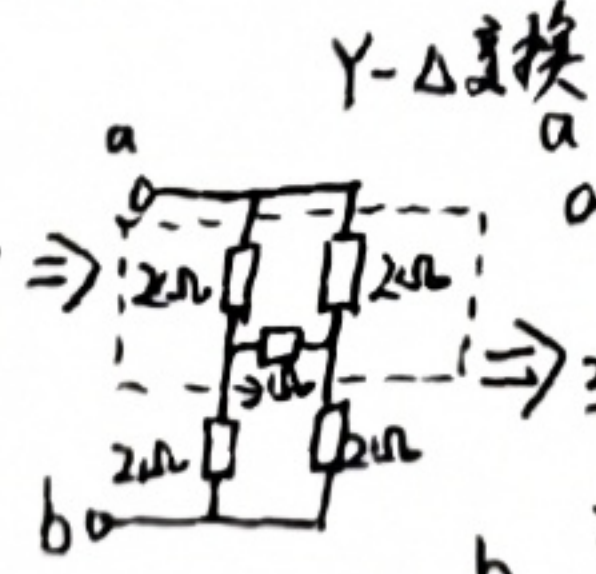
用网孔电流法求开路电压 U_{oc} .

1: $I_1 \cdot (2+2+2) - (2+2)I_2 = 4$
2: $I_2 \cdot (2+2+2+2) - (2+2)I_1 = 0$
 $\Rightarrow \begin{cases} I_1 = 1A \\ I_2 = 0.5A \end{cases}$

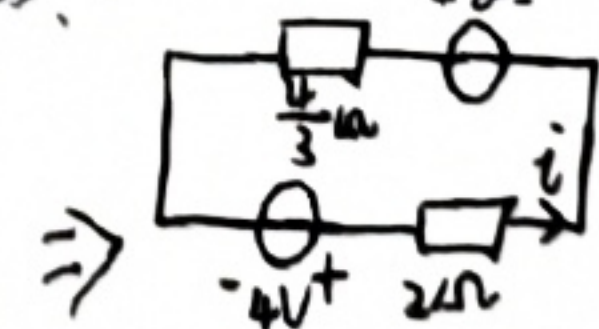
$U_{oc} = U_{ab} = I_2 \cdot (2+2) - I_1 \cdot 2 = 0$. (或电桥电路, $R_1 = R_2 = R_3 = R_4$, 即 $U_{ab} = 0$)

求短路电流:

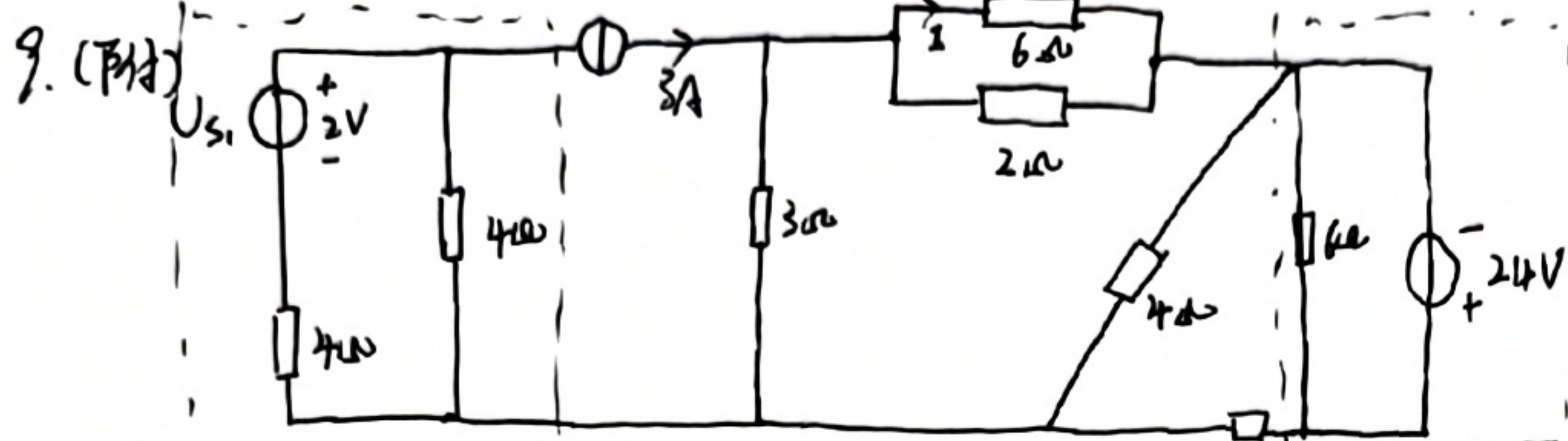
电源置 0.



$R_{eq} = \frac{2}{3} + \left[\frac{2}{3} + 2 \parallel \left(\frac{2}{3} + 2 \right) \parallel \left(\frac{2}{3} + 2 \right) \right]$
 $= \frac{4}{3} \Omega$

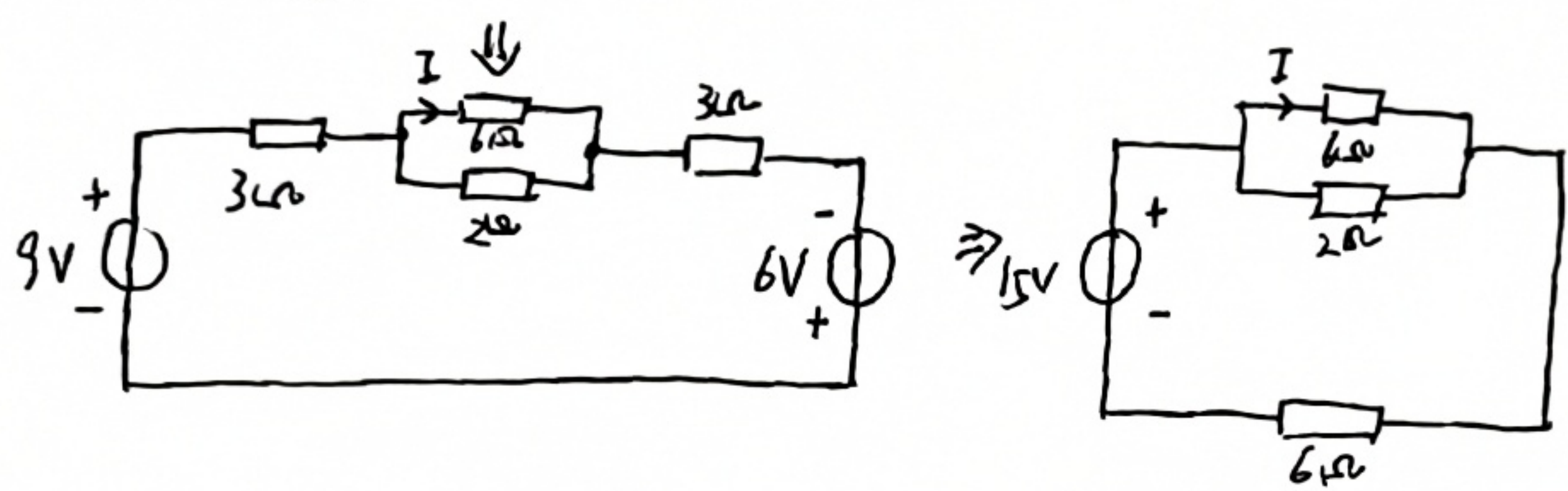
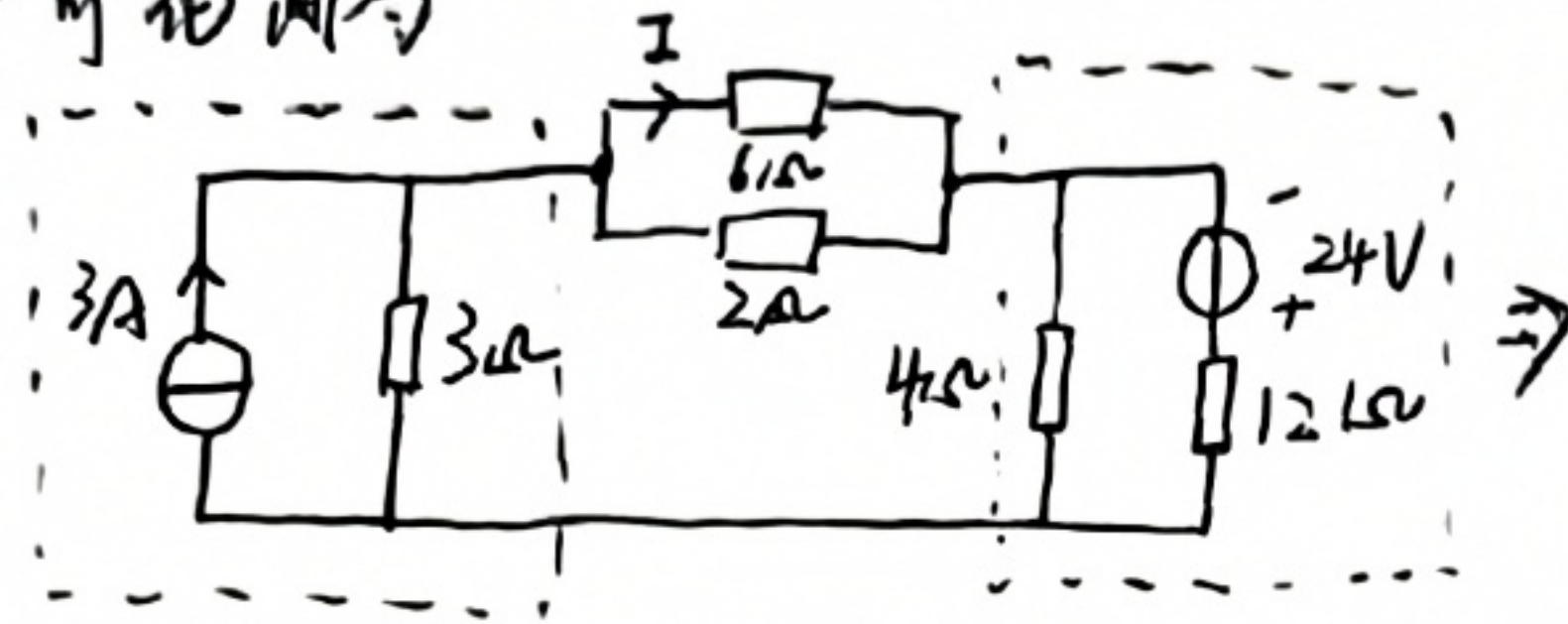


$i = \frac{4}{2 + \frac{4}{3}} A = 1.2A$

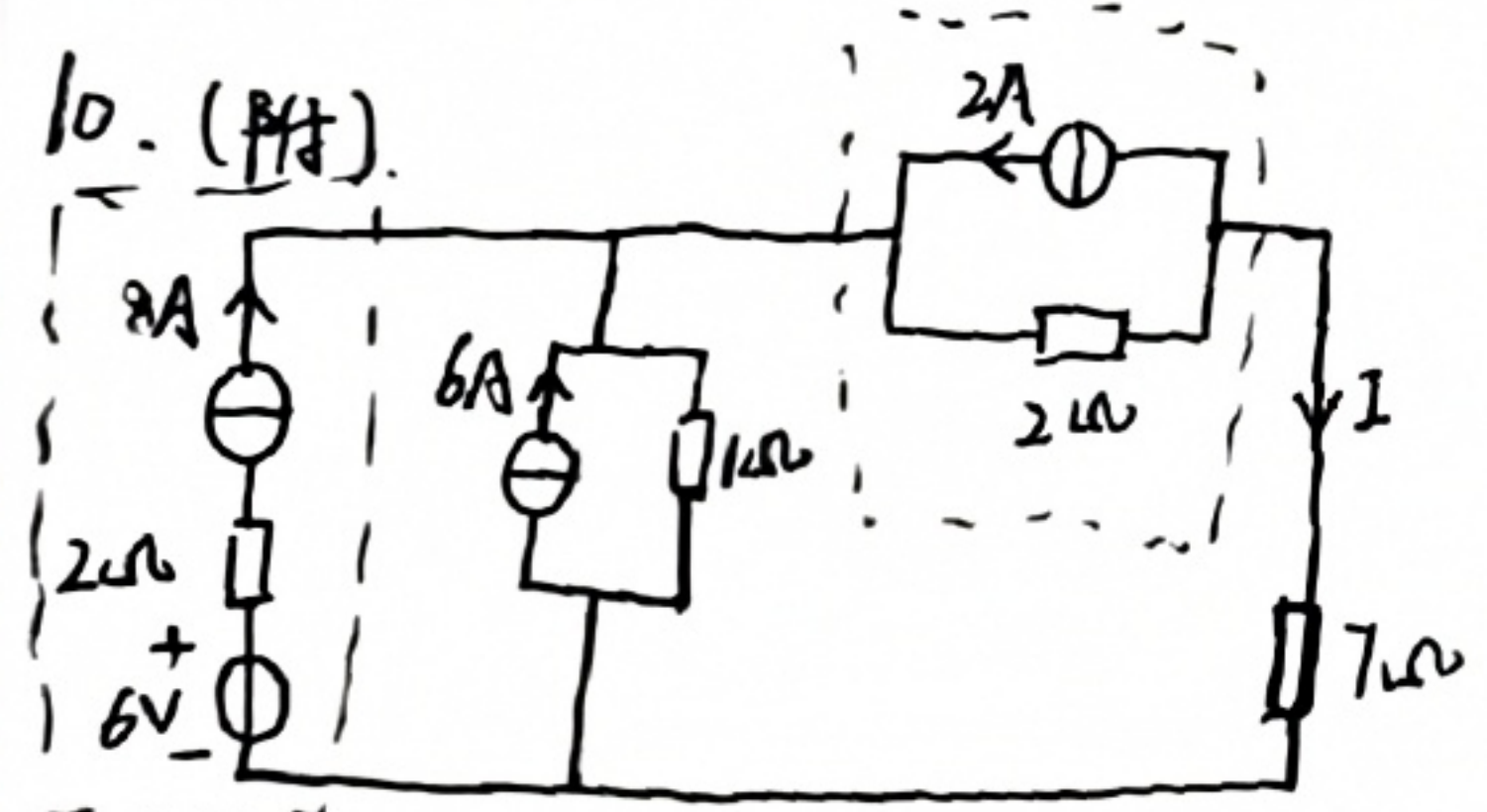


理想电流源, 阻值无限大, 可视为开路(断路), 即左边电路无作用.
理想电压源, 阻值无限小, 可视为短路(导线). 右边 6Ω 电阻被短路.

3p 电路可化简为

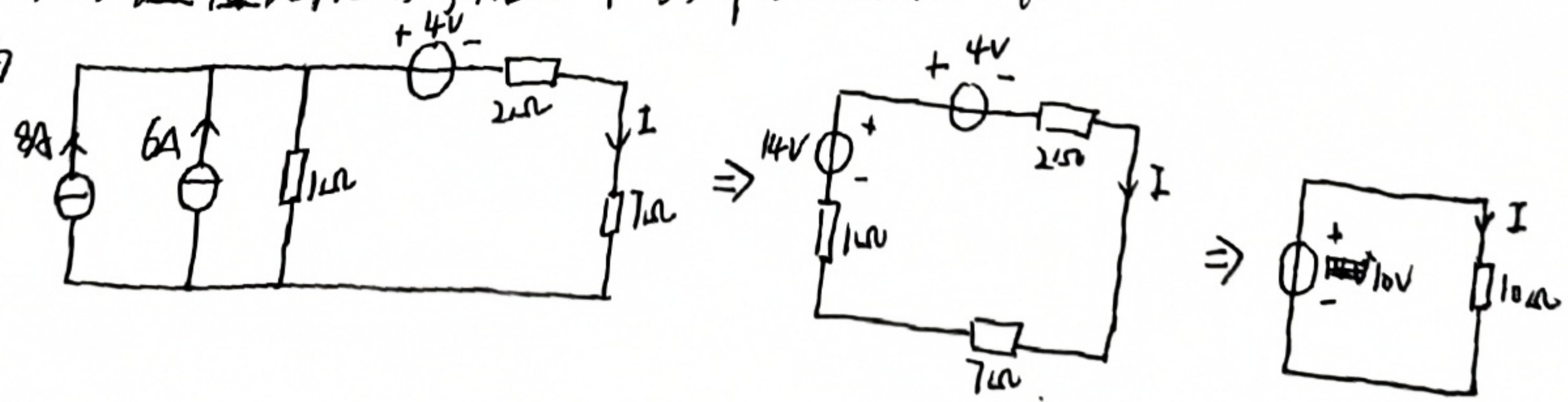


$$3p \quad I = \frac{15}{6 + 6//2} \times \frac{2}{6+2} A = 0.5 A.$$



理想电流源, 阻值无限大, 可视为开路, 即左边电路无用.

电路可化为



$$3p: I = \frac{10}{10} A = 1 A.$$